DO NOW — RECALL

# Recursion: base cases and returns

Name: _____ Class: _____ Date: _____

**1. A C# function is declared int Double(int n). What does return n * 2; do when n is 4?**

**2. Starting total = 1, repeat total = total * i for i = 1, 2, 3. What is total after each iteration?**

**3. Values A, B and C are pushed onto a stack in that order. Which leaves first when popping?**

**4. A function AddTwo(x) returns x + 2. Calculate AddTwo(AddTwo(3)) and show the inner and outer calls.**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

It sends the value 8 back to the caller. A return statement also ends the current invocation.

### 2. Model answer

1, 2 and 6. Each iteration multiplies the previous total by the new i.

### 3. Model answer

C. A stack is last in, first out (LIFO).

### 4. Model answer

The inner AddTwo(3) returns 5; then the outer AddTwo(5) returns 7.

### Vocabulary — know these terms

recursion — A function calling itself to solve a smaller problem.

base case — A condition that stops further recursive calls.

recursive case — A call to the same function with a smaller or simpler input.

call stack — Memory tracking active function calls and their local state.

return value — The result passed back to a caller.

NEW LEARNING

# Read, annotate and explain

### Start with a small input

Factorial(4) calls Factorial(3); it cannot finish multiplying until the inner call returns.

### Reach the stopping rule

Factorial(0) returns 1. It makes no further recursive call.

### Return through the calls

Factorial(1) = 1 × 1; Factorial(2) = 2 × 1; Factorial(3) = 3 × 2; Factorial(4) = 4 × 6 = 24.

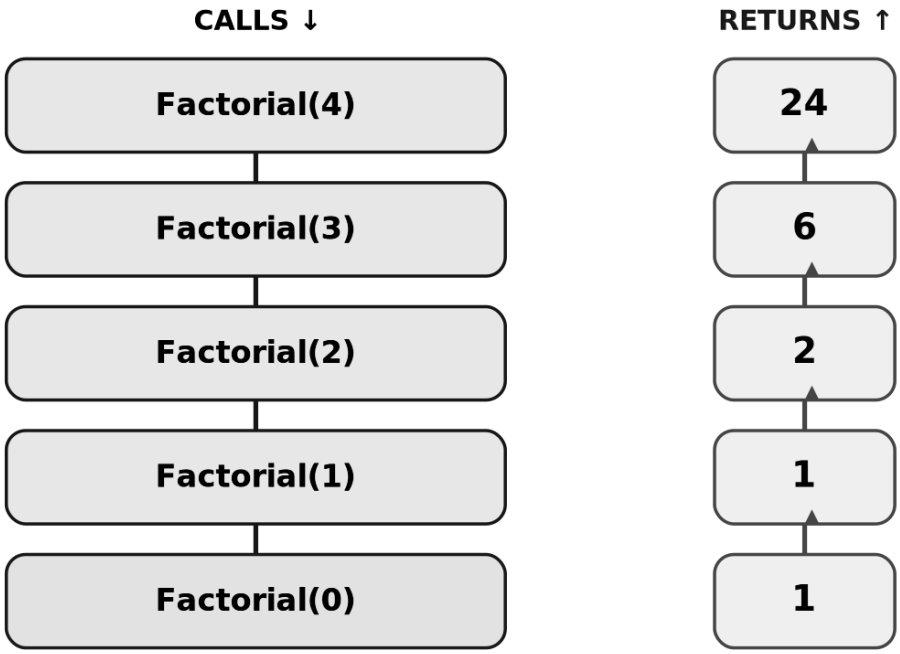
### Explain one step in your own words, using the correct technical terms.

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### What is the program actually doing?

Think: Why does the input get smaller? What stops another call?




Recursive calls moving downward and return values moving upwards. Factorial(4), Factorial(3), Factorial(2), Factorial(1), Factorial(0).

I DO — WORKED EXAMPLE

# Follow the worked method

**Trace Factorial(3). Show each call and its return value.**

### Calls

3 → 2 → 1 → 0. The argument decreases by one each time.

### Base case

Factorial(0) returns 1. No more calls occur.

### Returns

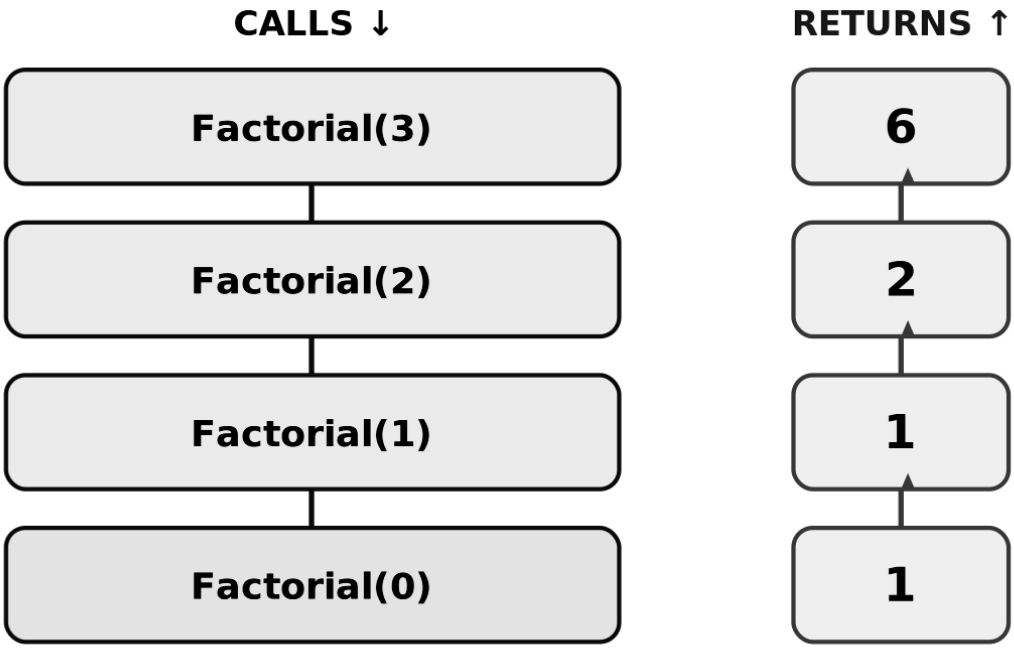
Factorial(1)=1, Factorial(2)=2, Factorial(3)=6. Final answer: 6.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Trace Factorial(3). Show each call and its return value.**

3 → 2 → 1 → 0. The argument decreases by one each time.




Call trace and return trace: Factorial(3); Factorial(2); Factorial(1); Factorial(0). Return values: 6, 2, 1, 1.

**WE DO — GUIDED PRACTICE**

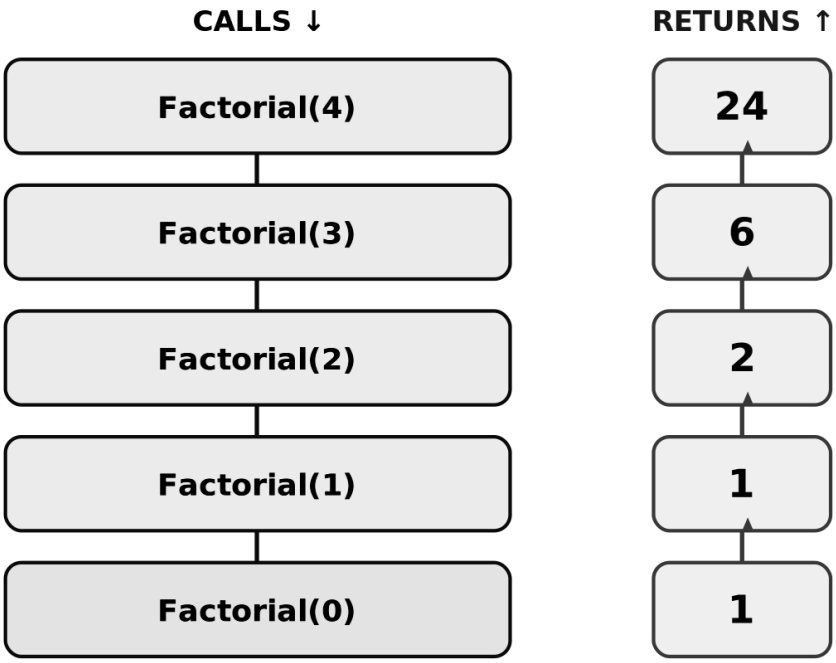
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Trace Factorial(4) without running the program. Identify the base case and final returned value.**

Factorial(4) → ... → Factorial(0)




Call trace and return trace: Factorial(4); Factorial(3); Factorial(2); Factorial(1); Factorial(0). Return values: 24, 6, 2, 1, 1.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Complete the calls and returns for Factorial(2).

### Question 2

What happens if Factorial has no base case? Explain using the call stack.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Modify Factorial to reject negative input; write the C# guard and explain the reason.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**If a recursive call reaches its base case, all previous calls have already returned.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.